

Realisation problem for the space of knots in the 3-sphere

OPG-37131

L1 - TRACTABLE

PARTIALLY SOLVED

Topology

ORIGINAL PROBLEM STATEMENT

Problem Given a link L in S^3 , let the symmetry group of L be denoted $Sym(L) = \pi_0 Diff(S^3, L)$ ie: isotopy classes of diffeomorphisms of S^3 which preserve L , where the isotopies are also required to preserve L .

Now let L be a hyperbolic link. Assume L has the further ‘Brunnian’ property that there exists a component L_0 of L such that $L \setminus L_0$ is the unlink. Let A_L be the subgroup of $Sym(L)$ consisting of diffeomorphisms of S^3 which preserve L_0 together with its orientation, and which preserve the orientation of S^3 .

There is a representation $A_L \rightarrow \pi_0 Diff(L \setminus L_0)$ given by restricting the diffeomorphism to the $L \setminus L_0$. It’s known that A_L is always a cyclic group. And $\pi_0 Diff(L \setminus L_0)$ is a signed symmetric group – the wreath product of a symmetric group with $\mathbb{Z}_2$.

Problem: What representations can be obtained?

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